

Experiment No - 2 :

Object — Determination of efficiency of DC shunt motor by load test.

Apparatus Required —

DC Shunt motor

Tachometer

DC ammeter

DC voltmeter

Load

Starter

Connecting wires.

Brief Theory —

The efficiency of DC shunt motor is given by —

$$\% \eta = \frac{\text{output power}}{\text{input power}} \times 100.$$

To measure the output power of motor, load is connected to the pulley.

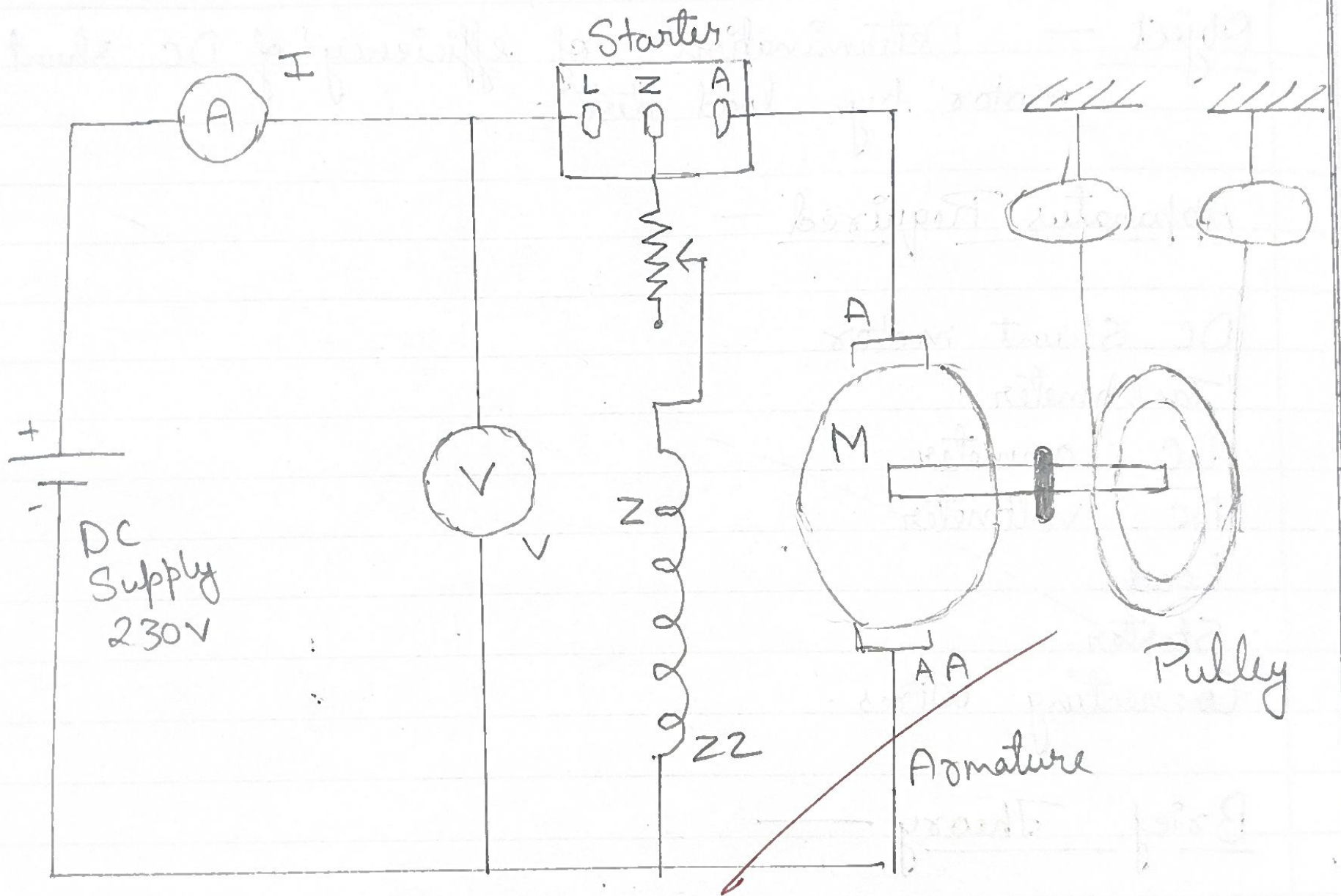


Figure .

Output power $W_2 = \omega T$ where $T = g_2 [S_1 - S_2]$ & $\omega = \frac{2\pi N}{60} \cdot [S_1 - S_2]$ is the difference b/w the readings of load, r is radius of pulley and N is the speed of motor. $g = 9.81 \text{ m/s}^2$

Input power $W_1 = VI$, where V & I are the readings of voltmeter & ammeter.

Procedure —

1. Connect the circuit as per figure.
2. Switch on the power supply.
3. Record the readings of voltmeter & ammeter and speed of motor at no-load condⁿ.
4. Slowly increase the load on water cooled pulley.
5. Take the readings of voltmeter, ammeter and record the corresponding speed.
6. Repeat the above step for different loads.



Observation Table —

S.No.	Reading of Voltmeter V	Reading of ammeter I	Input power $V I [W_1]$	Difference $[S_1 - S_2]$ Kg	Speed [RPM]	Output power W_2	% η $\frac{W_2 \times 100}{W_1}$
1.	199.7	2.337	466.7	3-1=2	1302	226.6	48.5
2.	196.8	3.117	613.4	5-1.5=3.5	1280	390.4	63.6
3.	197	3.886	765.5	7-2.5=4.5	1276	500.1	65.3

Average — $\frac{48.5 + 63.6 + 65.3}{3} = 59.1\%$

Calculation — $\% \eta = \frac{W_2}{W_1} \times 100$

Result —

The efficiency of dc shunt motor is 59.1%

Date 14/09/26Expt. No. 2Page No. 8Precautions —

1. All the connection should be tight.
2. All apparatus should be of suitable range and ratings.
3. Never touch the live terminals.
4. Under no-load condⁿ, switch off the input supply.

~~apsara~~
14/09/26

Teacher's Signature _____